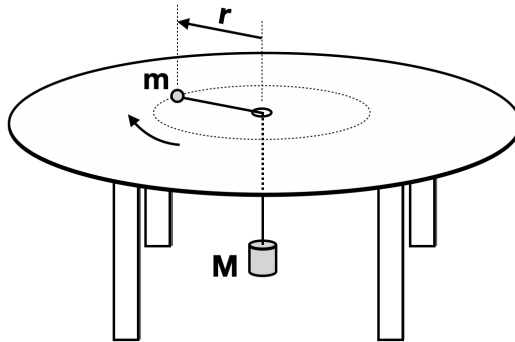


Physics I (Spring 2024): Midterm Exam Solution

Apr. 26, 2024

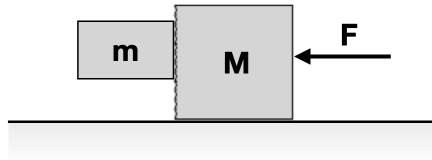
[total 25 pts, closed book, 90 minutes]

1. (a) [1 pt] Two forces, $\mathbf{F}_1 = (1.0 \text{ N})\hat{\mathbf{i}} + (1.0 \text{ N})\hat{\mathbf{j}} + (-2.0 \text{ N})\hat{\mathbf{k}}$ and $\mathbf{F}_2 = (2.0 \text{ N})\hat{\mathbf{i}} + (-1.0 \text{ N})\hat{\mathbf{j}} + (3.0 \text{ N})\hat{\mathbf{k}}$ are acting on an object. Find the total work done by the two forces if the object undergoes the displacement $\mathbf{D} = (1.0 \text{ m})\hat{\mathbf{j}} + (-5.0 \text{ m})\hat{\mathbf{k}}$.
- (b) [1 pt] A force $\mathbf{F} = (2.0 \text{ N})\hat{\mathbf{i}} + (-3.0 \text{ N})\hat{\mathbf{j}} + (1.0 \text{ N})\hat{\mathbf{k}}$ acts at the point $P = (1.0 \text{ m}, 5.0 \text{ m}, 2.0 \text{ m})$. Find the torque due to $\mathbf{F}$ about the origin.
- (c) [2 pt] A particle of mass $m = 1.2 \text{ kg}$ is connected to a cylinder of mass $M = 5.0 \text{ kg}$ by a massless string that passes through a small hole in the center of a table (see figure). The particle m slides in a circle of radius $r = 0.20 \text{ m}$ on the frictionless table which is $h = 1.0 \text{ m}$ above the ground. In the meantime, the cylinder M hangs under the table and does not move up or down. During the circular motion, the string always remains tight. Then, the string suddenly breaks and the particle m flies off the table horizontally. How far from the edge of the table horizontally does the particle m land?

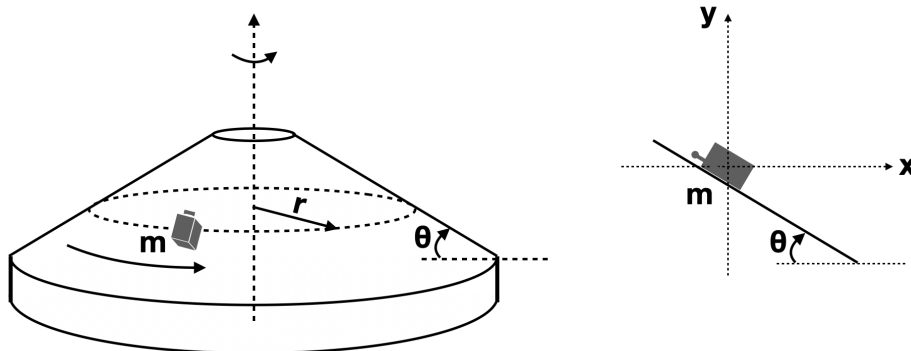


- (a) $W = -5 \text{ J}$.
- (b) $\boldsymbol{\tau} = (11 \text{ N} \cdot \text{m})\hat{\mathbf{i}} + (3.0 \text{ N} \cdot \text{m})\hat{\mathbf{j}} + (-13 \text{ N} \cdot \text{m})\hat{\mathbf{k}}$.
- (c) $Mg = \frac{mv_{\text{circ}}^2}{r} \rightarrow x = v_{\text{circ}} \sqrt{\frac{2h}{g}} = \sqrt{\frac{2Mrh}{m}}$.

2. (a) [2 pt] The two blocks in the figure, $m = 2.0 \text{ kg}$ and $M = 10 \text{ kg}$, are not attached to each other. The surface beneath the larger block is frictionless, but the coefficient of static friction between the blocks is $\mu_s = 0.24$ (see figure). Find the *minimum* magnitude of the horizontal force $\mathbf{F}$ required to keep the smaller block from slipping down the larger block.

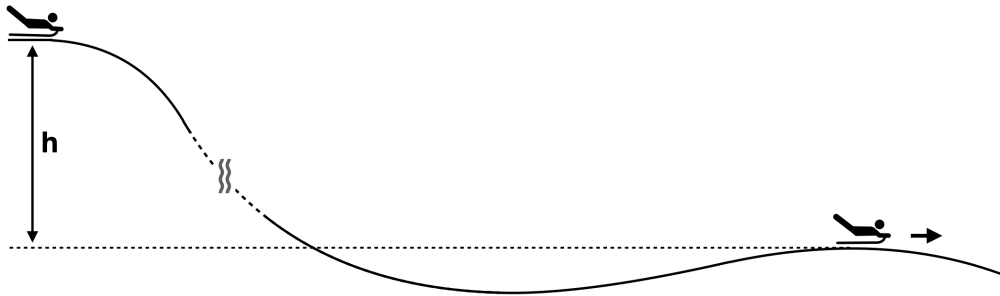


(b) [3 pt] An airport baggage carousel can be modeled as a large cone that rotates steadily about its vertical axis. Its metallic slope makes an angle $\theta = 30^\circ$ with the horizontal (see figure). (i) A suitcase of mass $m = 30 \text{ kg}$ is placed on the carousel at a position $r = 7.5 \text{ m}$ measured horizontally from the rotation axis. The suitcase goes around once every 40 seconds without sliding down the slope. Calculate the force of static friction exerted by the carousel on the suitcase. (ii) Later, the drive motor malfunctions and turns the carousel at a higher speed, while the suitcase is bumped to a different position, $r' = 8.0 \text{ m}$ from the rotation axis. Now, with the carousel rotating once every 32 seconds, the suitcase is on the verge of sliding down the slope. Find the coefficient of static friction, μ_s , between the suitcase and the carousel. (Note: You may find it useful to complete the free-body diagram below showing all the forces acting on the suitcase.)



- (a) For the two blocks moving together, $F - F' = Ma$ for block M and $F' = ma$ for block m , where F' is the force exerted on m by M . From these you obviously acquire $a = \frac{F}{M+m}$. Also, the minimum value of F' can be acquired from $\mu_s F'_{\min} = mg$. Combining the equations, you get $F'_{\min} = \frac{mg}{\mu_s} = ma_{\min} = F_{\min} \cdot \frac{m}{M+m} \rightarrow F_{\min} = \frac{(M+m)g}{\mu_s}$.
- (b-1) With $v = \frac{2\pi r}{T}$, the centripetal force in the negative x -direction is $F_{\text{cen}} = \frac{mv^2}{r} = f_s \cos \theta - F_N \sin \theta$, while there is no net force in the y -direction, i.e., $0 = f_s \sin \theta + F_N \cos \theta - mg$. One can easily find $f_s = \frac{mv^2}{r} \cdot \cos \theta + mg \sin \theta \simeq 150 \text{ N}$.
- (b-2) With $v' = \frac{2\pi r'}{T'}$, the centripetal force in the negative x -direction is $F'_{\text{cen}} = \frac{mv'^2}{r'} = f_{s, \max} \cos \theta - F'_N \sin \theta = F'_N (\mu_s \cos \theta - \sin \theta)$, while in the y -direction $0 = f_{s, \max} \sin \theta + F'_N \cos \theta - mg = F'_N (\mu_s \sin \theta + \cos \theta) - mg$. The two equations yield $F'_N = -\frac{mv'^2}{r'} \cdot \sin \theta + mg \cos \theta \simeq 250 \text{ N}$, and $\mu_s = \frac{mg - F'_N \cos \theta}{F'_N \sin \theta} \simeq 0.62$.

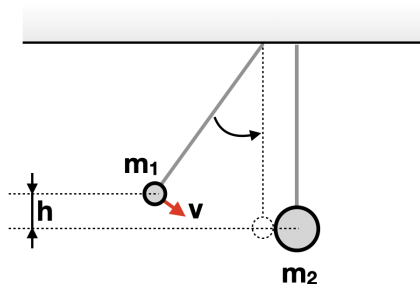
3. (a) [1 pt] A skeleton athlete starts from rest at the top of an icy hill. The track has sixteen turns and drops $h = 120\text{ m}$ in elevation from start to finish (see figure). The athlete reaches the end of the run with a speed $v = 39\text{ m/s}$. During the athlete's descent, how much work is done on the *athlete+sled* of combined mass $m = 110\text{ kg}$ by (i) the gravitational force, and (ii) the non-conservative forces, such as friction and air resistance? (Note: Assume that the speed at the beginning of the run is small and can be ignored. Pay special attention to the signs and significant digits of your answers.)



(b) [2 pt] We are at rest relative to an inertial reference frame, watching a rocket accelerate through deep space with no gravitational or drag forces acting on it. Let M be the mass of the rocket and v its velocity at an arbitrary time t . v_{rel} is the fuel's exhaust speed relative to the rocket, which we assume to be constant. For a rocket with a constant fuel consumption (exhaust) rate, show that the change in the rocket's speed is given by $\Delta v = v_{\text{rel}} \cdot \ln \frac{M_i}{M_f}$ as its mass changes from M_i to M_f .

- (a) $\Delta K = W = W_g + W_{\text{nc}} \rightarrow W_{\text{nc}} = \Delta K - W_g = \frac{1}{2}mv^2 - mgh = -4.6 \times 10^4\text{ J}$.
- (b) See Section 9-9 of Halliday & Resnick, or Section 9.11 of Thornton & Marion.

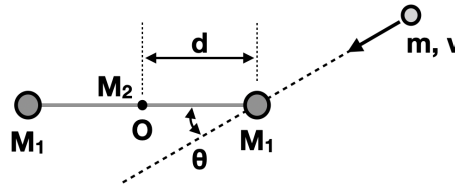
4. (a) [2 pt] Two steel balls are hanging from hinged massless rods that can swing without friction on the horizontal axle. Starting with an initial speed $v_0 = 5.0\text{ m/s}$ at a height of $h_0 = 0.40\text{ m}$, one of the balls ($m_1 = 1.5\text{ kg}$) swings downward and strikes another ball of ($m_2 = 4.5\text{ kg}$) that is at rest (see figure). Assuming that the collision is elastic, find how high each ball swings after the collision.



(b) [3 pt] Two small balls of mass $M_1 = 1.0\text{ kg}$ each are attached to the ends of a uniform thin rod of length $2d = 2.0\text{ m}$ and mass $M_2 = 0.60\text{ kg}$. (i) The rod is free to rotate in a horizontal plane without friction about a vertical axis through its center of mass, O (see figure). With the

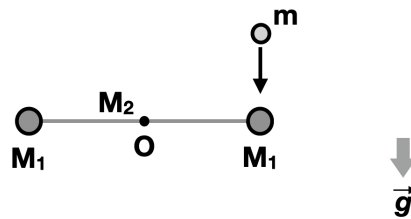
rod+ball system initially at rest, a lump of clay ($m = 0.20 \text{ kg}$) is thrown to one of the balls, hitting it with a speed of $v = 3.0 \text{ m/s}$ (in this horizontal plane) at an angle $\theta = 30^\circ$ and then sticking to it, as shown in the figure. What is the angular speed of the system after the clay hits? (Note: The table of various rotational inertias is in the last page of this exam.)

(i)



(ii) In another scenario, the same rod is now free to rotate in a *vertical* plane without friction about a *horizontal* axis through its center of mass, O (see figure). With the *rod+ball* system initially horizontal and at rest, a lump of clay ($m = 0.20 \text{ kg}$) vertically falls onto one of the balls, hitting it with a speed of $v = 3.0 \text{ m/s}$ (in this vertical plane) and then sticking to it. What is the ratio of the kinetic energy of the system just after the collision to that of the clay just before the collision? Then, after rotating through an angle ϕ , the *rod+ball+clay* system momentarily stops. Determine $\sin \phi$. (Note: One may find that $\sin \phi$ is negative.)

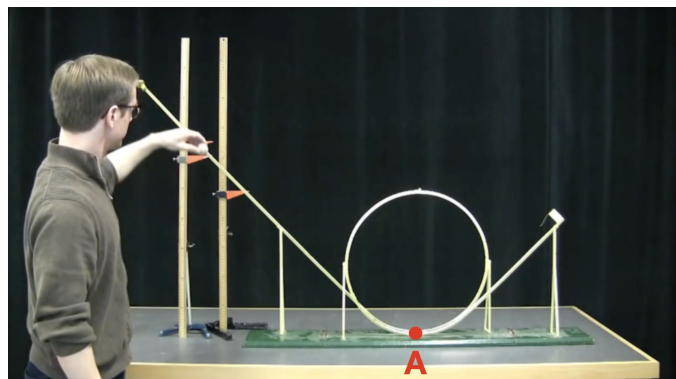
(ii)



- (a) $v_{1i} = \sqrt{v_0^2 + 2gh_0}$. Then, from Halliday & Resnick Eqs.(9-67)-(9-68), $v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i}$ and $v_{2f} = \frac{2m_1}{m_1 + m_2} v_{1i}$.
- (b-1) $I_{\text{before}} = 2M_1 d^2 + \frac{1}{12} M_2 (2d)^2 \rightarrow I_{\text{after}} = 2M_1 d^2 + \frac{1}{12} M_2 (2d)^2 + m d^2 = 2.4 \text{ kg m}^2$, while $L_{\text{before}} = m v d \sin \theta = 0.30 \text{ kg m}^2/\text{s} = I_{\text{after}} \cdot \omega_1$. Hence, $\omega_1 = 0.13 \text{ rad/s}$.
- (b-2) $L_{\text{before}} = m v d = 0.60 \text{ kg m}^2/\text{s} = L_{\text{after}} = I_{\text{after}} \cdot \omega_2$, which gives $\omega_2 = 0.25 \text{ rad/s}$ and $\frac{E_{K, \text{after}}}{E_{K, \text{before}}} = \frac{I_{\text{after}} \omega_2^2}{m v^2} \simeq 0.083$.
- (b-3) Since the *rod+ball* system is symmetric with respect to O regardless of the rotation angle ϕ , the rotational kinetic energy of the *rod+ball+clay* system must be completely converted to the potential energy of the clay when the system momentarily stops after rotating through ϕ ($> \pi$). Thus, $E_{K, \text{after}} = \frac{1}{2} I_{\text{after}} \omega_2^2 = m g d \sin(\phi - \pi) \rightarrow \sin \phi = -\sin(\phi - \pi) = -\frac{I_{\text{after}} \omega_2^2}{2 m g d} \simeq -0.038$.

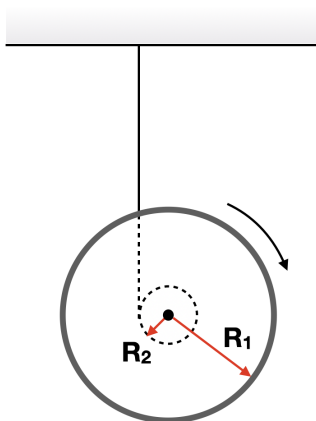
5. (a) [2 pt] Recall the video we watched in the class (see figure). Assume that a uniform solid ball of mass m starts from the rest and rolls smoothly down the track without slipping. The radius of the ball, r , is very small compared to the radius of the circular loop, R . The work done by rolling friction is negligible. (i) What is the minimum initial height h (in units of R)

for which the ball will remain in contact with the track when reaching the top of the loop? (ii) When released from h you just found, what is the normal force on the ball from the track (in units of mg) as it passes through the bottom of the loop (point A; see figure)?



(iii) Now, suppose that the ball is squeezed to a sphere of radius $\frac{r}{2}$ while maintaining its mass m . What is the minimum initial height h (in units of R) for which the ball will make a complete loop-the-loop on the circular part of the track? Explain how your answer differs from the one in (i). (Note: The table of various rotational inertias is in the last page of this exam.)

(b) [2 pt] A yo-yo can be modeled as a system of two uniform thin disks of mass $\frac{1}{2}M$ and radius R_1 connected by a massless axle of radius R_2 in between. The thickness of the axle can be neglected. A massless string is suspended vertically from a ceiling, and the other end is wrapped multiple times around the axle. When the yo-yo is released from the rest vertically down, it rotates as the string unwinds. Show that the yo-yo's linear acceleration is $a = g \left(1 + \frac{R_1^2}{2R_2^2} \right)^{-1}$.

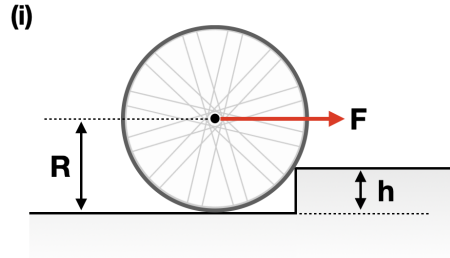


- (a-1) At the top of the loop, the centripetal force is $F_{\text{cen},T} = F_{N,T} - mg = -\frac{mv_T^2}{R}$, from which one can find that, on the verge of losing contact with the track ($F_{N,T} = 0$), the shell's speed must be $v_T = \sqrt{gR}$. Then, $mgh = mg(2R) + \frac{1}{2}mv_T^2 + \frac{1}{2}\left(\frac{2}{5}mr^2\right)\left(\frac{v_T}{r}\right)^2 = mg(2R) + \frac{7}{10}mv_T^2 = \frac{27}{10}mgR$ yields $h = 2.7R$. At point A, $mg\left(\frac{27}{10}R\right) = \frac{7}{10}mv_A^2$ gives $F_{N,A} = F_{\text{cen},A} + mg = \frac{mv_A^2}{R} + mg = \frac{34}{7}mg$ where the positive sign indicates an upward direction (towards the center of the circle).

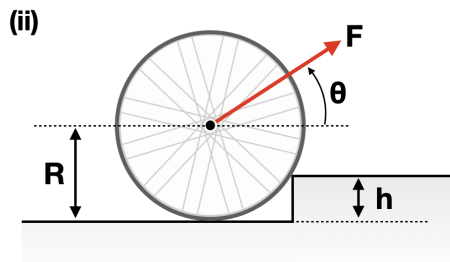
- (a-2) With r replaced with $\frac{r}{2}$, one finds $mgh = mg(2R) + \frac{1}{2}mv_T^2 + \frac{1}{2}\left(\frac{2}{5}m\left(\frac{r}{2}\right)^2\right)\left(\frac{v_T}{r/2}\right)^2 = \frac{27}{10}mgR$, which yields the same answer $h = 2.7R$.

- (b) $Ma = Mg - T$ and $R_2T = \left(\frac{1}{2}MR_1^2\right)\left(\frac{a}{R_2}\right)$. Or, see Section 11-3 of Halliday & Resnick.

6. (a) [2 pt] A titanium wheel of mass $m = 6.0$ kg and radius $R = 20$ cm is resting against a step of height $h = 10$ cm. (i) A horizontal force $\vec{F}$ is applied at the axle of the wheel to roll it over the step obstacle (see figure). What is the minimum force magnitude necessary? (Note: As the magnitude of $\vec{F}$ increases, there comes a moment when the wheel begins to rise up and loses contact with the ground.)



(ii) Now, in a different scenario, a constant force $\vec{F}$ is applied at the axle of the wheel at angle θ above the horizontal, just enough to start lifting the wheel off the ground (see figure). What is the value of θ that requires the least magnitude of the force $\vec{F}$? At this angle θ you just found, what is the magnitude of $\vec{F}$?



(b) [2 pt] In class, we discussed several examples where the physics principles are used to understand the movement of an athlete or even a celestial body. Among the following sets of keywords, select *four* and craft a concise yet physically meaningful narrative for each set that incorporates all the keywords in it. For each set, a paragraph of at least 3-4 sentences is expected to explain how the physics principle you learned in the class is employed to explain an amazing athletic performance, or how it provides analogies for the complex celestial event in the distant universe. To aid the grader, circle all the keywords in your answers. Use diagrams if desired.

- High jumper (높이뛰기 선수): *center of mass*, jumper's arching body, projectile motion
- Spinning figure skater: *angular momentum*, skater's rotational inertia, isolated system
- Spinning neutron star: *angular momentum*, collapsing core's rotational inertia, pulsar
- Supernova explosion: *momentum transfer*, imploding shells of matter, exclusion principle
- Exoplanet detection: *Doppler effect*, binary system, invisible object, redshift and blueshift

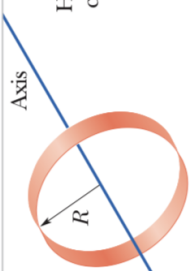
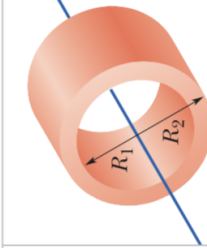
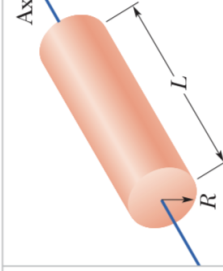
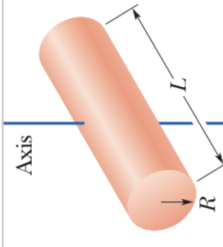
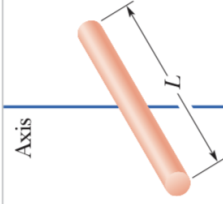
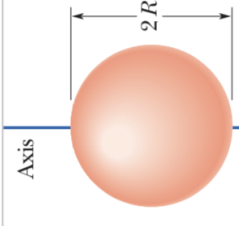
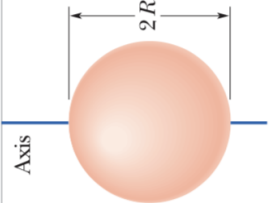
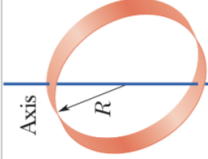
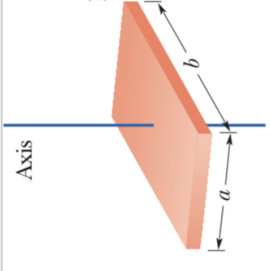
• (a-1) The requirements for equilibrium are $F_{E,x} + F = 0$, $F_{E,y} - mg = 0$, and $-FR \sin 150^\circ + mgR \sin 120^\circ = 0$, where F_E is the force from the step edge towards the wheel's center. Either the third equation or the first and second equations combined gives $F = \frac{mg \cos 30^\circ}{\sin 30^\circ} \simeq 102$ N.

• (a-2) The requirements for equilibrium are $F_{E,x} + F \cos \theta = 0$, $F_{E,y} + F \sin \theta - mg = 0$, and $-FR \sin(150^\circ - \theta) + mgR \sin 120^\circ = 0$. The third equation transforms to $F = \frac{mg \sin 120^\circ}{\sin(150^\circ - \theta)} = \frac{mg \cos 30^\circ}{\sin(30^\circ + \theta)} \geq mg \cos 30^\circ$, where the equality is when $\theta = 60^\circ$. Combined, the first and second

equations can also yield the same inequality by way of $\frac{F_{E,y}}{F_{E,x}} = \frac{mg - F \sin \theta}{-F \cos \theta} = -\tan 30^\circ$. At $\theta = 60^\circ$, the minimum force magnitude needed to roll the wheel over the step is $F = mg \cos 30^\circ \simeq 51 \text{ N}$.

- (b) See the class slides for Lectures 2-1, 5-1, 5-2, 7-1 and 7-2 that include the examples listed here.

Table 10-2 Some Rotational Inertias

 Hoop about central axis Axis R $I = MR^2$ (a)	 Annular cylinder (or ring) about central axis Axis R_1 R_2 $I = \frac{1}{2}M(R_1^2 + R_2^2)$ (b)	 Solid cylinder (or disk) about central axis Axis L R $I = \frac{1}{2}MR^2$ (c)
 Solid cylinder (or disk) about central diameter Axis L R $I = \frac{1}{4}MR^2 + \frac{1}{12}ML^2$ (d)	 Thin rod about axis through center perpendicular to length Axis L $I = \frac{1}{12}ML^2$ (e)	 Solid sphere about any diameter Axis $2R$ $I = \frac{2}{5}MR^2$ (f)
 Thin spherical shell about any diameter Axis $2R$ $I = \frac{2}{3}MR^2$ (g)	 Hoop about any diameter Axis R $I = \frac{1}{2}MR^2$ (h)	 Slab about perpendicular axis through center Axis a b $I = \frac{1}{12}M(a^2 + b^2)$ (i)